DSA0503 – Query Processing for Data Science

CO2 - AT1 - Database Design and Connectivity

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1. Vehicle Registration Database (SQLite)

Task:

- Create a SQLite database named `vehicle_registration.db`.
- Create a `Vehicle` table with Registration Number, Owner Name, Vehicle Type, Brand, Model, and Year.
- Insert five sample vehicle records.
- Search for a vehicle using its Registration Number.
- Display the retrieved vehicle details.

Implementation

Code:

```
# --------------------------------------------
# Vehicle Registration Database using SQLite
# --------------------------------------------

import sqlite3

# Connect to SQLite database
conn = sqlite3.connect("vehicle_registration.db")

# Create cursor object
cursor = conn.cursor()

# Create Vehicle table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Vehicle(
    RegistrationNumber TEXT PRIMARY KEY,
    OwnerName TEXT NOT NULL,
    VehicleType TEXT NOT NULL,
```

```python
    Brand TEXT NOT NULL,
    Model TEXT NOT NULL,
    Year INTEGER NOT NULL
)
""")

print("Vehicle table created successfully.\n")

# Delete existing records (optional to avoid duplicates)
cursor.execute("DELETE FROM Vehicle")

# Insert 5 vehicle records
vehicles = [
    ("TN10AB1234", "Arun Kumar", "Car", "Hyundai", "i20", 2023),
    ("TN22CD5678", "Priya", "Bike", "Honda", "Shine", 2022),
    ("TN07EF9012", "Rahul", "Car", "Maruti", "Swift", 2021),
    ("TN15GH3456", "Divya", "Scooter", "TVS", "Jupiter", 2024),
    ("TN09IJ7890", "Karthik", "Car", "Toyota", "Innova", 2020)
]

cursor.executemany("INSERT INTO Vehicle VALUES(?,?,?,?,?,?)", vehicles)

# Save records
conn.commit()

print("5 Vehicle records inserted successfully.\n")

# Search using Registration Number
reg_no = input("Enter Registration Number to Search: ")

cursor.execute("SELECT * FROM Vehicle WHERE RegistrationNumber=?", (reg_no,))

result = cursor.fetchone()

# Display Result
if result:
    print("\nVehicle Found")
    print("-------------------------------")
    print("Registration Number :", result[0])
    print("Owner Name          :", result[1])
    print("Vehicle Type        :", result[2])
    print("Brand               :", result[3])
    print("Model               :", result[4])
    print("Year                :", result[5])
else:
    print("\nVehicle Not Found.")

# Close connection
```

```python
conn.close()
```

Output:

```
5 Vehicle records inserted successfully.

Enter Registration Number to Search: TN10AB1234

Vehicle Found
-------------------------------
Registration Number : TN10AB1234
Owner Name          : Arun Kumar
Vehicle Type        : Car
Brand               : Hyundai
Model               : i20
Year                : 2023
```

2. Employee Payroll Management System (MySQL)

Task:

- Connect Python to the MySQL payroll database.
- Create an Employee table with Employee ID, Name, Department, Salary, and Designation.
- Insert five sample employee records.
- Update the salary of a specified employee using Employee ID.
- Display the updated employee details.

Implementation

Code:

```
import mysql.connector

# Connect to MySQL Database
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root123",   # Replace with your MySQL password
    database="payroll"
)

cursor = conn.cursor()

# Create Employee table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Employee(
    EmployeeID INT PRIMARY KEY,
    Name VARCHAR(50),
    Department VARCHAR(50),
    Salary FLOAT,
    Designation VARCHAR(50)
)
""")

print("Employee table created successfully.")
```

```python
# Delete existing records (optional)
cursor.execute("DELETE FROM Employee")

# Insert 5 employee records
employees = [
    (101, "Divyesh", "AI & DS", 50000, "Software Engineer"),
    (102, "Arun", "HR", 40000, "HR Executive"),
    (103, "Priya", "Finance", 45000, "Accountant"),
    (104, "Rahul", "Marketing", 42000, "Marketing Executive"),
    (105, "Karthik", "IT", 55000, "System Administrator")
]

insert_query = """
INSERT INTO Employee
(EmployeeID, Name, Department, Salary, Designation)
VALUES (%s, %s, %s, %s, %s)
"""

cursor.executemany(insert_query, employees)
conn.commit()

print("5 Employee records inserted successfully.")

# Update salary
emp_id = int(input("Enter Employee ID to update salary: "))
new_salary = float(input("Enter New Salary: "))

update_query = """
UPDATE Employee
SET Salary = %s
WHERE EmployeeID = %s
"""

cursor.execute(update_query, (new_salary, emp_id))
conn.commit()

print("Salary updated successfully.\n")

# Display updated employee details
cursor.execute(
    "SELECT * FROM Employee WHERE EmployeeID=%s",
    (emp_id,)
)

employee = cursor.fetchone()

if employee:
```

```python
    print("Updated Employee Details")
    print("---------------------------")
    print("Employee ID :", employee[0])
    print("Name        :", employee[1])
    print("Department  :", employee[2])
    print("Salary      :", employee[3])
    print("Designation :", employee[4])
else:
    print("Employee not found.")

conn.close()
```

Output:

```
Enter Employee ID to update salary: 101
Enter New Salary: 60000
Salary updated successfully.

Updated Employee Details
---------------------------
Employee ID : 101
Name        : Divyesh
Department  : AI & DS
Salary      : 60000.0
Designation : Software Engineer
```